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### Camera optical lens

#### Abstract

A camera optical lens includes, from an object side to an image side, a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens. At least one of the first lens to the eighth lens has a free-form surface, and the camera optical lens satisfies:  $-3.50 \leq f2/f1 \leq -1.50$ ; and  $-2.00 \leq R9/R10 \leq -0.80$ , where  $f1$  is a focal length of the first lens,  $f2$  is a focal length of the second lens,  $R9$  is a central curvature radius of an object side surface of the fifth lens, and  $R10$  is a central curvature radius of an image side surface of the fifth lens. The camera optical lens has a large aperture, a wide angle, and ultra-thinness, as well as excellent optical performance.

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## Background/Summary

### TECHNICAL FIELD

(1) The present disclosure relates to the field of optical lenses, and more particularly, to a camera optical lens suitable for portable terminal devices such as smart phones and digital cameras, and suitable for imaging devices such as monitors and PC lenses.

### BACKGROUND

(2) With development of camera lenses, higher and higher requirements are put forward for imaging of the lens. The “night scene photography” and “background blur” of the lens have also become important indicators to measure an imaging of the lens. The structures in related art have insufficient focal power distribution, lens spacing and lens shape settings, resulting in insufficient ultra-thin and wide-angle lenses. Moreover, the rotationally symmetric aspherical surface cannot correct aberrations well. A free-form surface is a non-rotationally symmetric surface, which can better balance aberrations and improve the imaging quality; besides, processing of the free-form surface has been gradually mature. With the increasing requirements for imaging of the lens, it is very important to provide a free-curve surface in the design of a lens, especially in the design of a wide-angle and ultra-wide-angle lens.

### SUMMARY

(3) In view of the above-mentioned problems, a purpose of the present disclosure is to provide a camera optical lens, which has a large aperture, a wide angle, and ultra-thinness, as well as excellent optical performance.

(4) A camera optical lens is provided and the camera optical lens includes, from an object side to an image side, a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens. At least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens, the sixth lens, the seventh lens, or the eighth lens has a free-form surface, and the camera optical lens satisfies:

$$-3.50 \leq f_2/f_1 \leq -1.50; \text{ and}$$

$$-2.00 \leq R_9/R_{10} \leq -0.80,$$

(5) where  $f_1$  denotes a focal length of the first lens,  $f_2$  denotes a focal length of the second lens,  $R_9$  denotes a central curvature radius of an object side surface of the fifth lens, and  $R_{10}$  denotes a central curvature radius of an image side surface of the fifth lens.

(6) As an improvement, the camera optical lens satisfies:

$$1.40 \leq d_{10}/d_{12} \leq 3.00,$$

(7) where  $d_{10}$  denotes an on-axis distance from the image side surface of the fifth lens to an object side surface of the sixth lens, and  $d_{12}$  denotes an on-axis distance from an image side surface of the sixth lens to an object side surface of the seventh lens.

(8) As an improvement, the camera optical lens satisfies:

$$-4.43 \leq f_1/f \leq -1.24;$$

$$-2.24 \leq (R_1+R_2)/(R_1-R_2) \leq -0.28; \text{ and}$$

$$0.05 \leq d_1/TTL \leq 0.14,$$

(9) where  $f$  denotes a focal length of the camera optical lens,  $R_1$  denotes a central curvature radius of an object side surface of the first lens,  $R_2$  denotes a central curvature radius of an image side surface of the first lens,  $d_1$  denotes an on-axis thickness of the first lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(10) As an improvement, the camera optical lens satisfies:

$$1.81 \leq f_2/f \leq 10.68;$$

$$-11.57 \leq (R_3+R_4)/(R_3-R_4) \leq -2.79; \text{ and}$$

$$0.03 \leq d_3/TTL \leq 0.10,$$

(11) where  $f$  denotes a focal length of the camera optical lens,  $R_3$  denotes a central curvature radius of an object side surface of the second lens,  $R_4$  denotes a central curvature radius of an image side surface of the second lens,  $d_3$  denotes an on-axis thickness of the second lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(12) As an improvement, the camera optical lens satisfies:

$$-48.31 \leq f_3/f \leq 3.71;$$

$$-1.57 \leq (R_5+R_6)/(R_5-R_6) < 15.62; \text{ and}$$

$$0.02 \leq d_5/TTL \leq 0.12,$$

(13) where  $f$  denotes a focal length of the camera optical lens,  $f_3$  denotes a focal length of the third lens,  $R_5$  denotes a central curvature radius of an object side surface of the third lens,  $R_6$  denotes a central curvature radius of an image side surface of the third lens,  $d_5$  denotes an on-axis thickness of the third lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(14) As an improvement, the camera optical lens satisfies:

$$0.56 \leq f_4/f \leq 3.19;$$

$$0.01 \leq (R_7+R_8)/(R_7-R_8) \leq 1.51; \text{ and}$$

$0.04 \leq d7/TTL \leq 0.19$ ,

(15) where  $f$  denotes a focal length of the camera optical lens,  $f4$  denotes a focal length of the fourth lens,  $R7$  denotes a central curvature radius of an object side surface of the fourth lens,  $R8$  denotes a central curvature radius of an image side surface of the fourth lens,  $d7$  denotes an on-axis thickness of the fourth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(16) As an improvement, the camera optical lens satisfies:

$-6.34 \leq f5/f \leq -1.96$ ;

$-0.06 \leq (R9+R10)/(R9-R10) \leq 0.47$ ;

$0.02 \leq d9/TTL \leq 0.07$ ,

(17) where  $f$  denotes a focal length of the camera optical lens,  $f5$  denotes a focal length of the fifth lens,  $R9$  denotes a central curvature radius of the object side surface of the fifth lens,  $R10$  denotes a central curvature radius of the image side surface of the fifth lens,  $d9$  denotes an on-axis thickness of the fifth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(18) As an improvement, the camera optical lens satisfies:

$-11.90 \leq f6/f \leq -1.72$ ;

$-0.71 \leq (R11+R12)/(R11-R12) \leq -0.16$ ; and

$0.04 \leq d11/TTL \leq 0.13$ ,

(19) where  $f$  denotes a focal length of the camera optical lens,  $f6$  denotes a focal length of the sixth lens,  $R11$  denotes a central curvature radius of an object side surface of the sixth lens,  $R12$  denotes a central curvature radius of an image side surface of the sixth lens,  $d11$  denotes an on-axis thickness of the sixth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(20) As an improvement, the camera optical lens satisfies:

$0.42 \leq f7/f \leq 1.38$ ;

$0.25 \leq (R13+R14)/(R13-R14) \leq 1.05$ ; and

$0.04 \leq d13/TTL \leq 0.12$ ,

(21) where  $f$  denotes a focal length of the camera optical lens,  $f7$  denotes a focal length of the seventh lens,  $R13$  denotes a central curvature radius of an object side surface of the seventh lens,  $R14$  denotes a central curvature radius of an image side surface of the seventh lens,  $d13$  denotes an on-axis thickness of the seventh lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(22) As an improvement, the camera optical lens satisfies:

$-2.63 \leq f8/f \leq -0.85$ ;

$1.14 \leq (R15+R16)/(R15-R16) \leq 3.83$ ; and

$0.03 \leq d15/TTL \leq 0.11$ ,

(23) where  $f$  denotes a focal length of the camera optical lens,  $f8$  denotes a focal length of the eighth lens,  $R15$  denotes a central curvature radius of an object side surface of the eighth lens,  $R16$  denotes a central curvature radius of an image side surface of the eighth lens,  $d15$  denotes an on-axis thickness of the eighth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(24) With the camera optical lens of the present disclosure, the lens has good optical performance with a large aperture, ultra-thinness and a wide angle. Meanwhile, at least one of the first lens to the eighth lens has a free-form surface, thereby effectively correcting aberration and improving the performance of the optical system. It is suitable for mobile phone camera lens assembly and WEB camera lens composed of imaging elements for high pixel such as high-pixel CCD and CMOS.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) Many aspects of the exemplary embodiment can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

(2) FIG. 1 is a schematic structural diagram of a camera optical lens according to Embodiment 1 of the present disclosure;

(3) FIG. 2 illustrates a situation where RMS spot diameter of the camera optical lens shown in FIG. 1 is located in a first quadrant;

(4) FIG. 3 is a schematic structural diagram of a camera optical lens according to Embodiment 2 of the present disclosure;

(5) FIG. 4 illustrates a situation where RMS spot diameter of the camera optical lens shown in FIG. 3 is located in a first quadrant;

(6) FIG. 5 is a schematic structural diagram of a camera optical lens according to Embodiment 3 of the present disclosure; and

(7) FIG. 6 illustrates a situation where RMS spot diameter of the camera optical lens shown in FIG. 5 is within a first quadrant.

### DESCRIPTION OF EMBODIMENTS

(8) In order to better illustrate the purpose, technical solutions and advantages of the present disclosure, the embodiments of the present disclosure will be described in details as follows with reference to the accompanying drawings. However, it should be understood by those skilled in the art that, technical details are set forth in the embodiments of the present disclosure so as to better illustrate the present disclosure. However, the technical solutions claimed in the present disclosure can be achieved without these technical details and various changes and modifications based on the following embodiments.

#### Embodiment 1

(9) With reference to the accompanying drawings, the present disclosure provides a camera optical lens 10. FIG. 1 illustrates a

camera optical lens **10** according to Embodiment 1 of the present disclosure. The camera optical lens **10** includes eight lenses. Specifically, the camera optical lens **10** includes, from an object side to an image side, a first lens L1, a second lens L2, an aperture S **1**, a third lens L3, a fourth lens L4, a fifth lens L5, a sixth lens L6, a seventh lens L7, and an eighth lens L8. Optical elements such as an optical filter GF can be provided between the eighth lens L8 and an image plane Si.

(10) As an example, the first lens L1 is made of a plastic material, the second lens L2 is made of a plastic material, the third lens L3 is made of a plastic material, the fourth lens L4 is made of a plastic material, the fifth lens L5 is made of a plastic material, the sixth lens L6 is made of a plastic material, the seventh lens L7 is made of a plastic material, and the eighth lens L8 is made of a plastic material. In other embodiments, each lens can be made of another material.

(11) As an example, at least one of the first lens L1 to the eighth lens L8 includes a free-form surface, and the free-form surface contributes to correction of aberrations such as astigmatism, field curvature, and distortion in a wide-angle optical system.

(12) As an example, a focal length of the first lens L1 is  $f_1$ , and a focal length of the second lens L2 is  $f_2$ , then the camera optical lens satisfies the following condition:  $-3.50 \leq f_2/f_1 \leq -1.50$ , which specifies a ratio of the focal length of the second lens L2 to the focal length of the first lens L1. With this condition, it is beneficial to improving the imaging quality.

(13) As an example, a central curvature radius of an object side surface of the fifth lens L5 is  $R_9$ , and a central curvature radius of an image side surface of the fifth lens L5 is  $R_{10}$ , then the camera optical lens satisfies the following condition:  $-2.00 \leq R_9/R_{10} \leq -0.80$ , which specifies a shape of the fifth lens L5. Within a range defined by this condition, it is beneficial to reducing a degree of light deflection and improving the image quality.

(14) When the camera optical lens **10** of the present disclosure includes at least one free-form surface, and the focal length of the related lens and the central curvature radius of the related lens satisfy the above-mentioned condition, the camera optical lens **10** can have high performance and meet the requirements of a large aperture, a wide angle and ultra-thinness.

(15) As an example, an on-axis distance from an image side surface of the fifth lens L5 to an object side surface of the sixth lens L6 is  $d_{10}$ , and an on-axis distance from an image side surface of the sixth lens L6 to an object side surface of the seventh lens L7 is  $d_{12}$ , then the camera optical lens satisfies the following condition:  $1.40 \leq d_{10}/d_{12} < 3.00$ . With this condition, the field curvature of the system can be balanced, and the imaging quality can be improved.

(16) As an example, the first lens L1 has a negative refractive power, an object side surface of the first lens L1 is concave in a paraxial region, and an image side surface of the first lens L1 is concave in a paraxial region. In other embodiments, the first lens L1 can have a positive refractive power.

(17) As an example, a focal length of the camera optical lens **10** is  $f$ , and the camera optical lens satisfies the following condition:  $-4.43 \leq f_1/f \leq -1.24$ , which limits a ratio of the focal length of the first lens L1 to the focal length  $f$ . By controlling negative refractive power of the first lens L1 within a reasonable range, it is beneficial to correcting aberration of the optical system. As an example, the camera optical lens satisfies the following condition:  $-2.77 \leq f_1/f \leq -1.55$ .

(18) As an example, a central curvature radius of the object side surface of the first lens L1 is  $R_1$ , and a central curvature radius of the image side of the first lens L1 is  $R_2$ , then the camera optical lens satisfies the following condition:  $-2.24 \leq (R_1+R_2)/(R_1-R_2) \leq -0.28$ . By reasonably controlling a shape of the first lens L1, the first lens L1 can effectively correct spherical aberration of the system. As an example, the camera optical lens satisfies the following condition:  $-1.40 \leq (R_1+R_2)/(R_1-R_2) \leq -0.35$ .

(19) As an example, an on-axis thickness of the first lens L1 is  $d_1$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.05 \leq d_1/TTL \leq 0.14$ , which is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.07 \leq d_1/TTL \leq 0.11$ .

(20) As an example, the second lens L2 has a positive refractive power, the object side surface of the second lens L2 is convex in a paraxial region, and the image side surface of the second lens L2 is concave in a paraxial region. In other embodiments, the second lens L2 can have a negative refractive power.

(21) As an example, the camera optical lens satisfies the following condition:  $1.81 \leq f_2/f \leq 10.68$ , which specifies a ratio of the focal length of the second lens L2 to the focal length  $f$ . With this condition, the second lens L2 has an appropriate positive refractive power, which is beneficial to reducing aberration of the system, as well as beneficial to achieving ultra-thinness and a wide angle of the lens. As an example, the camera optical lens satisfies the following condition:  $2.89 \leq f_2/f \leq 8.54$ .

(22) As an example, a central curvature radius of the object side surface of the second lens L2 is  $R_3$ , and a central curvature radius of the image side surface of the second lens L2 is  $R_4$ , then the camera optical lens satisfies the following condition:  $-11.57 \leq (R_3+R_4)/(R_3-R_4) \leq -2.79$ , which specifies a shape of the second lens L2. With this condition, with development of ultra-thinness and wide-angle of the lens, it is beneficial to correcting on-axis color aberration. As an example, the camera optical lens satisfies the following condition:  $-7.23 \leq (R_3+R_4)/(R_3-R_4) \leq -3.49$ .

(23) As an example, an on-axis thickness of the second lens L2 is  $d_3$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.03 \leq d_3/TTL \leq 0.10$ . With this condition, it is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.04 \leq d_3/TTL \leq 0.08$ .

(24) As an example, the third lens L3 has a positive refractive power, the object side surface of the third lens L3 is convex in a paraxial region, and the image side surface of the third lens L3 is convex in a paraxial region. In other embodiments, the third lens L3 can have a negative refractive power.

(25) As an example, a focal length of the third lens L3 is  $f_3$ , then the camera optical lens satisfies the following condition:  $-48.31 \leq f_3/f \leq 3.71$ . With this condition, limitation on the third lens L3 with this condition can effectively smooth the light angle of the camera lens and reduce tolerance sensitivity. As an example, the camera optical lens satisfies the following condition:  $-30.19 \leq f_3/f \leq 2.97$ .

(26) As an example, a central curvature radius of the object side surface of the third lens L3 is  $R_5$ , and a central curvature radius of the image side surface of the third lens L3 is  $R_6$ , then the camera optical lens satisfies the following condition:

$-1.57 \leq (R5+R6)/(R5-R6) \leq 15.62$ , which specifies a shape of the third lens L3. Within a range defined by this condition and the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of an off-axis angle. As an example, the camera optical lens satisfies the following condition:  $-0.98 \leq (R5+R6)/(R5-R6) \leq 12.50$ .

(27) As an example, an on-axis thickness of the third lens L3 is  $d5$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.02 \leq d5/TTL \leq 0.12$ . With this condition, it is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.04 \leq d5/TTL \leq 0.09$ .

(28) As an example, the fourth lens L4 has a positive refractive power, the object side surface of the fourth lens L4 is concave in a paraxial region, and the image side surface is convex in a paraxial region. In other embodiments, the fourth lens L4 can have a negative refractive power.

(29) As an example, the focal length of the fourth lens L4 is  $f4$ , then the camera optical lens satisfies the following condition:  $0.56 \leq f4/f \leq 3.19$ , which specifies a ratio of the focal length of the fourth lens L4 to the focal length  $f$ . With this condition, it is beneficial to improving the performance of the optical system. As an example, the camera optical lens satisfies the following condition:  $0.89 \leq f4/f \leq 2.55$ .

(30) As an example, a central curvature radius of the object side surface of the fourth lens L4 is  $R7$ , and a central curvature radius of the image side surface of the fourth lens L4 is  $R8$ , then the camera optical lens satisfies the following condition:  $0.01 \leq (R7+R8)/(R7-R8) \leq 1.51$ , which specifies a shape of the fourth lens L4. With this condition and the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of an off-axis angle. As an example, the camera optical lens satisfies the following condition:  $0.02 \leq (R7+R8)/(R7-R8) \leq 1.21$ .

(31) As an example, an on-axis thickness of the fourth lens L4 is  $d7$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.04 \leq d7/TTL \leq 0.19$ . With this condition, it is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.06 \leq d7/TTL \leq 0.15$ .

(32) As an example, the fifth lens L5 has a negative refractive power, the object side surface of the fifth lens L5 is concave in a paraxial region, and the image side surface of the fifth lens L5 is concave in a paraxial region. In other embodiments, the fifth lens L5 can have a positive refractive power.

(33) As an example, a focal length of the fifth lens L5 is  $f5$ , then the camera optical lens satisfies the following condition:  $-6.34 \leq f5/f \leq -1.96$ , which specifies a ratio of the focal length of the fifth lens L5 to the focal length. With this condition, it is beneficial to improving the performance of the optical system. As an example, the camera optical lens satisfies the following condition:  $-3.96 \leq f5/f \leq -2.45$ .

(34) As an example, a central curvature radius of the object side surface of the fifth lens L5 is  $R9$ , and a central curvature radius of the image side surface of the fifth lens L5 is  $R10$ , then the camera optical lens satisfies the following condition:  $-0.06 \leq (R9+R10)/(R9-R10) \leq 0.47$ , which specifies a shape of the fifth lens L5. With this condition and the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of an off-axis angle. As an example, the camera optical lens satisfies the following condition:  $-0.04 \leq (R9+R10)/(R9-R10) \leq 0.38$ .

(35) As an example, an on-axis thickness of the fifth lens L5 is  $d9$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.02 \leq d9/TTL \leq 0.07$ . With this condition, it is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.03 \leq d9/TTL \leq 0.05$ .

(36) As an example, the sixth lens L6 has a negative refractive power, the object side surface of the sixth lens L6 is concave in a paraxial region, and the image side surface is concave in a paraxial region. In other embodiments, the sixth lens L6 can have a positive refractive power.

(37) As an example, a focal length of the sixth lens L6 is  $f6$ , then the camera optical lens satisfies the following condition:  $-11.90 \leq f6/f \leq -1.72$ . Reasonable distribution of the focal power enables the system to have better imaging quality and lower sensitivity. As an example, the camera optical lens satisfies the following condition:  $-7.44 \leq f6/f \leq -2.15$ .

(38) As an example, a central curvature radius of the object side surface of the sixth lens L6 is  $R11$ , and a central curvature radius of the image side surface of the sixth lens L6 is  $R12$ , then the camera optical lens satisfies the following condition:  $-0.71 \leq (R11+R12)/(R11-R12) \leq -0.16$ , which specifies a shape of the sixth lens L6. With this condition and the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of an off-axis angle. As an example, the camera optical lens satisfies the following condition:  $-0.44 \leq (R11+R12)/(R11-R12) \leq -0.20$ .

(39) As an example, an on-axis thickness of the sixth lens L6 is  $d11$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.04 \leq d11/TTL \leq 0.13$ . With this condition, it is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.07 \leq d11/TTL \leq 0.11$ .

(40) As an example, the seventh lens L7 has a positive refractive power, the object side surface of the seventh lens L7 is convex in a paraxial region, and the image side surface of the seventh lens L7 is convex in a paraxial region. In other embodiments, the seventh lens L7 can have a negative refractive power.

(41) As an example, a focal length of the seventh lens L7 is  $f7$ , then the camera optical lens satisfies the following condition:  $0.42 \leq f7/f \leq 1.38$ . Reasonable distribution of the focal power enables the system to have better imaging quality and lower sensitivity. As an example, the camera optical lens satisfies the following condition:  $0.67 \leq f7/f \leq 1.10$ .

(42) As an example, a central curvature radius of the object side surface of the seventh lens L7 is  $R13$ , and a central curvature radius of the image side surface of the seventh lens L7 is  $R14$ , then the camera optical lens satisfies the following condition:  $0.25 \leq (R13+R14)/(R13-R14) \leq 1.05$ , which specifies a shape of the seventh lens L7. With this condition and the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of an off-axis angle. As an example, the camera optical lens satisfies the following condition:  $0.41 \leq (R13+R14)/(R13-R14) \leq 0.84$ .

(43) As an example, an on-axis thickness of the seventh lens L7 is  $d_{13}$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.04 \leq d_{13}/TTL \leq 0.12$ . With this condition, it is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.06 \leq d_{13}/TTL \leq 0.10$ .

(44) As an example, the eighth lens L8 has a negative refractive power, the object side surface of the eighth lens L8 is convex in a paraxial region, and the image side surface is concave in a paraxial region. In other embodiments, the eighth lens L8 can have a positive refractive power.

(45) As an example, a focal length of the eighth lens L8 is  $f_8$ , then the camera optical lens satisfies the following condition:  $-2.63 \leq f_8/f \leq -0.85$ . Reasonable distribution of the focal power enables the system to have better imaging quality and lower sensitivity. As an example, the camera optical lens satisfies the following condition:  $-1.65 \leq f_8/f \leq -1.06$ .

(46) As an example, a central curvature radius of the object side surface of the eighth lens L8 is  $R_{15}$ , and a central curvature radius of the image side surface of the eighth lens L8 is  $R_{16}$ , then the camera optical lens satisfies the following condition:  $1.14 \leq (R_{15}+R_{16})/(R_{15}-R_{16}) \leq 3.83$ , which specifies a shape of the eighth lens L8. With this condition and the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of an off-axis angle. As an example, the camera optical lens satisfies the following condition:  $1.83 \leq (R_{15}+R_{16})/(R_{15}-R_{16}) \leq 3.06$ .

(47) As an example, an on-axis thickness of the eighth lens L8 is  $d_{15}$ , and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis, then the camera optical lens satisfies the following condition:  $0.03 \leq d_{15}/TTL \leq 0.11$ . With this condition, it is beneficial to achieving ultra-thinness. As an example, the camera optical lens satisfies the following condition:  $0.05 \leq d_{15}/TTL \leq 0.09$ .

(48) As an example, an F number FNO of the camera optical lens **10** is smaller than or equal to 1.85. Large aperture leads to good imaging performance. As an example, it is satisfied that the aperture value FNO is smaller than or equal to 1.82.

(49) As an example, a ratio of the optical length TTL of the camera optical lens **10** to a full FOV image height IH (in a diagonal direction) is smaller than or equal to 1.15, which is beneficial to achieving ultra-thinness, for example, smaller than or equal to 1.10. The field of view (FOV) in the diagonal direction is larger than or equal to  $110^\circ$ , which is beneficial to achieving a wide angle. As an example, FOV is larger than or equal to  $118^\circ$ .

(50) When the above-mentioned condition is satisfied, the camera optical lens **10** has good optical performance, and when the free-form surface is adopted, the designed image plane area can be matched with an actual use area, thereby improving the image quality of the effective area to the greatest extent; and according to the characteristics of the camera optical lens **10**, the camera optical lens **10** is suitable for a mobile phone camera lens assembly and a WEB camera lens composed of imaging elements for high pixels such as CCD and CMOS. The camera optical lens **10** of the present disclosure will be described in the following by examples. The reference signs described in each example are as follows. The unit of each of the focal length, the on-axis distance, the central curvature radius and the on-axis thickness is mm.

(51) TTL: the optical length (an on-axis distance from the object side surface of the first lens L1 to the image plane Si), in a unit of mm.

(52) FNO: a ratio of an effective focal length of the camera optical lens **10** to an entrance pupil diameter.

(53) Table 1, Table 2, and Table 3 show design data of the camera optical lens **10** according to the Embodiment 1 of the present disclosure. Herein, the object side surface and image side surface of the eighth lens L8 are free-form surfaces.

(54) TABLE-US-00001  
TABLE 1  
R d Nd Nd S1  $\infty$  d0 = -2.080 R1 -2.685 d1 = 0.589 nd1 1.5444 v1 56.43 R2 6.479 d2 = 0.784 R3 1.853 d3 = 0.317 nd2 1.6610 v2 20.53 R4 3.017 d4 = 0.307 R5 2.710 d5 = 0.470 nd3 1.5444 v3 56.43 R6 -22.486 d6 = 0.095 R7 -861.494 d7 = 0.486 nd4 1.5444 v4 56.43 R8 -1.875 d8 = 0.068 R9 -11.503 d9 = 0.276 nd5 1.6800 v5 18.40 R10 5.995 d10 = 0.151 R11 -3.812 d11 = 0.536 nd6 1.5444 v6 56.43 R12 7.993 d12 = 0.062 R13 3.212 d13 = 0.477 nd7 1.5444 v7 56.43 R14 -1.049 d14 = 0.045 R15 1.822 d15 = 0.461 nd8 1.6032 v8 28.29 R16 0.713 d16 = 0.600 R17  $\infty$  d17 = 0.210 ndg 1.5168 vg 64.17 R18  $\infty$  d18 = 0.217  
Herein, the representation of each reference sign is as follows. S1: aperture; R: curvature radius at a center of an optical surface; R1: central curvature radius of an object side surface of a first lens L1; R2: central curvature radius of an image side surface of the first lens L1; R3: central curvature radius of an object side surface of a second lens L2; R4: central curvature radius of an image side surface of the second lens L2; R5: central curvature radius of an object side surface of a third lens L3; R6: central curvature radius of an image side surface of the third lens L3; R7: central curvature radius of an object side surface of a fourth lens L4; R8: central curvature radius of an image side surface of the fourth lens L4; R9: central curvature radius of an object side surface of a fifth lens L5; R10: central curvature radius of an image side surface of the fifth lens L5; R11: central curvature radius of an object side surface of a sixth lens L6; R12: central curvature radius of an image side surface of the sixth lens L6; R13: central curvature radius of an object side surface of a seventh lens L7; R14: central curvature radius of an image side surface of the seventh lens L7; R15: central curvature radius of an object side surface of an eighth lens L8; R16: central curvature radius of an image side surface of the eighth lens L8; R17: central curvature radius of an object side surface of an optical filter GF; R18: central curvature radius of an image side surface of the optical filter GF; d: on-axis thickness of the lens, and on-axis distance between lenses; d0: on-axis distance from the aperture S1 to the object side surface of the first lens L1; d1: on-axis thickness of the first lens L1; d2: on-axis distance from the image side surface of the first lens L1 to the object side surface of the second lens L2; d3: on-axis thickness of the second lens L2; d4: on-axis distance from the image side surface of the second lens L2 to the object side surface of the third lens L3; d5: on-axis thickness of the third lens L3; d6: on-axis distance from the image side surface of the third lens L3 to the object side surface of the fourth lens L4; d7: on-axis thickness of the fourth lens L4; d8: on-axis distance from the image side surface of the fourth lens L4 to the object side surface of the fifth lens L5; d9: on-axis thickness of the fifth lens L5; d10: on-axis distance from the image side surface of the fifth lens L5 to the object side surface of the sixth lens L6; d11: on-axis thickness of the sixth lens L6; d12: on-axis distance from the image side surface of the sixth lens L6 to the object side surface of the seventh lens L7; d13: on-axis thickness of the seventh lens L7; d14: on-axis distance from the image side surface of the seventh lens L7 to the object side surface of the eighth lens L8; d15: on-axis thickness of the eighth lens L8; d16: on-axis distance from the image side surface of the eighth lens L8 to the object side surface of the optical filter GF;

d17: on-axis thickness of optical filter GF; d18: on-axis distance from the image side surface of the optical filter GF to the image plane; nd: refractive index of d-line; nd1: refractive index of d-line of the first lens L1; nd2: refractive index of d-line of the second lens L2; nd3: refractive index of d-line of the third lens L3; nd4: refractive index of d-line of the fourth lens L4; nd5: refractive index of d-line of the fifth lens L5; nd6: refractive index of d-line of the sixth lens L6; nd7: refractive index of d-line of the seventh lens L7; nd8: refractive index of d-line of the eighth lens L8; ndg: refractive index of d-line of the optical filter GF; vd: abbe number; v1: abbe number of the first lens L1; v2: abbe number of the second lens L2; v3: abbe number of the third lens L3; v4: abbe number of the fourth lens L4; v5: abbe number of the fifth lens L5; v6: abbe number of the sixth lens L6; v7: abbe number of the seventh lens L7; v8: abbe number of the eighth lens L8; and vg: abbe number of the optical filter GF.

(55) Table 2 shows aspherical data of each lens of the camera optical lens **10** according to the Embodiment 1 of the present disclosure.

(56) TABLE-US-00002 TABLE 2 Conic coefficient Aspherical coefficient k A4 A6 A8 A10 A12 R1 -2.5000E+01 7.9612E-02 -4.1699E-02 1.7283E-02 -5.2633E-03 1.1361E-03 R2 -9.1444E+00 2.6336E-01 -2.2263E-01 2.6504E-01 -3.0201E-01 2.8169E-01 R3 -6.0227E-01 6.7970E-02 1.1398E-01 -3.7014E-01 5.1257E-01 1.4861E+00 R4 9.9906E+00 1.1297E-01 -1.2790E-01 8.0135E-01 -1.5331E+00 8.4713E-01 R5 -2.8588E-01 4.8415E-02 -6.7193E-03 1.1140E-01 -2.8255E-01 4.5759E-01 R6 -1.0000E+01 -1.0430E-01 -1.5819E-01 3.3610E-01 -7.8443E-01 1.8484E+00 R7 -1.0000E+01 -7.6787E-02 -2.2557E-01 2.1795E-01 -4.2101E-01 1.2541E+00 R8 9.3006E-01 -5.2265E-02 -2.2630E-01 5.1495E-01 -1.5572E+00 3.4604E+00 R9 -6.0470E+00 -2.9794E-01 -7.7693E-02 -3.1516E-02 -2.8644E-01 1.4613E+00 R10 -1.3135E+00 -1.9874E-01 7.7362E-02 -9.1215E-02 1.0241E-01 8.0485E-02 R11 -4.4143E+00 -1.4755E-01 2.9974E-01 -5.9327E-01 8.5653E-01 -7.9668E-01 R12 8.3525E+00 -1.5640E-01 -8.8320E-01 1.4126E+00 -3.9805E-01 -1.3005E+00 R13 1.8899E+00 3.5307E-01 -1.0378E+00 1.6062E+00 -1.5206E+00 7.6668E-01 R14 -6.8961E-01 8.8212E-01 -9.1704E-01 1.2047E+00 -1.4594E+00 1.1516E+00 Conic coefficient Aspherical coefficient k A14 A16 A18 A20 / R1 -2.5000E+01 -1.6682E-04 1.5749E-05 -8.5993E-07 2.0669E-08 / R2 -9.1444E+00 -1.8892E-01 8.2584E-02 -2.0603E-02 2.1894E-03 / R3 -6.0227E-01 -6.5064E+00 1.0032E+01 -7.2237E+00 1.9792E+00 / R4 9.9906E+00 2.0425E+00 -2.5545E+00 0.0000E+00 0.0000E+00 / R5 -2.8588E-01 -3.2524E-01 0.0000E+00 0.0000E+00 0.0000E+00 / R6 -1.0000E+01 -1.4685E+00 1.4465E-01 0.0000E+00 0.0000E+00 / R7 -1.0000E+01 -9.0710E-01 8.4848E-02 0.0000E+00 0.0000E+00 / R8 9.3006E-01 -4.0398E+00 2.1694E+00 -2.7604E-01 0.0000E+00 / R9 -6.0470E+00 -2.0786E+00 9.8100E-01 0.0000E+00 0.0000E+00 / R10 -1.3135E+00 -1.2627E-01 3.6725E-02 0.0000E+00 0.0000E+00 / R11 -4.4143E+00 6.9633E-01 -5.3497E-01 2.4165E-01 -4.4109E-02 / R12 8.3525E+00 1.9551E+00 -1.2874E+00 4.2756E-01 -5.7761E-02 / R13 1.8899E+00 -1.3067E-01 -5.1382E-02 2.7170E-02 -3.5256E-03 / R14 -6.8961E-01 -5.6054E-01 1.6365E-01 -2.6266E-02 1.7813E-03 /

(57)

$$z = (cr^2) / \{1 + [1 - (k + 1)(c^2 r^2)]^{1/2}\} + A 4r^4 + A 6r^6 + A 8r^8 + A 10r^{10} + A 12r^{12} + A 14r^{14} + A 16r^{16} + A 18r^{18} + A 20r^{20}, \quad (1)$$

(58) where k represents a conic coefficient, A4, A6, A8, A10, A12, A14, A16, A18, and A20 represent aspherical coefficients, c represents the curvature at the center of the optical surface, r represents a vertical distance between a point on an aspherical curve and the optical axis, Z represents an aspherical depth (a vertical distance between a point on an aspherical surface, having a distance of r from the optic axis, and a surface tangent to a vertex of the aspherical surface on the optic axis).

(59) For convenience, the aspherical surface of each lens adopts the aspherical surface shown in the above equation (1). However, the present disclosure is not limited to the aspherical surface defined by the polynomial form expressed by the equation (1).

(60) Table 3 shows free-form surface data in the camera optical lens **10** according to the Embodiment 1 of the present disclosure.

(61) TABLE-US-00003 TABLE 3 Free-form surface coefficient k X.sup.4Y.sup.0 X.sup.2Y.sup.2 X.sup.0Y.sup.4 X.sup.6Y.sup.0 X.sup.4Y.sup.2 X.sup.2Y.sup.4 X.sup.0Y.sup.6 R15 -1.2917E+00 -1.2645E-01 -2.5261E-01 -1.2760E-01 -6.2044E-01 -1.8645E+00 -1.8617E+00 -6.1863E-01 R16 -3.7641E+00 -1.9148E-01 -3.8219E-01 -1.9294E-01 1.0472E-01 3.1362E-01 3.1362E-01 1.0694E-01 X.sup.8Y.sup.0 X.sup.6Y.sup.2 X.sup.4Y.sup.4 X.sup.2Y.sup.6 X.sup.0Y.sup.8 X.sup.10Y.sup.0 X.sup.8Y.sup.2 X.sup.6Y.sup.4 R15 1.2137E+00 4.8587E+00 7.2898E+00 4.8544E+00 1.2132E+00 -1.1097E+00 -5.5494E+00 -1.1100E+01 R16 -1.8738E-02 -7.5054E-02 -1.1122E-01 -7.4967E-02 -1.9844E-02 -1.5590E-02 -7.7850E-02 -1.5538E-01 X.sup.4Y.sup.6 X.sup.2Y.sup.8 X.sup.0Y.sup.10 X.sup.12Y.sup.0 X.sup.10Y.sup.2 X.sup.8Y.sup.4 X.sup.6Y.sup.6 X.sup.6Y.sup.6 R15 -1.1101E+01 -5.5494E+00 -1.1102E+00 5.8130E-01 3.4873E+00 8.7177E+00 1.1625E+01 8.7189E+00 R16 -1.5634E-01 -7.8145E-02 -1.5574E-02 1.2179E-02 7.3082E-02 1.8245E-01 2.4335E-01 1.8250E-01 X.sup.4Y.sup.8 X.sup.2Y.sup.10 X.sup.0Y.sup.12 X.sup.14Y.sup.0 X.sup.12Y.sup.2 X.sup.10Y.sup.4 X.sup.8Y.sup.6 X.sup.6Y.sup.8 R15 3.4890E+00 5.8124E-01 -1.8071E-01 -1.2652E+00 -3.7953E+00 -6.3256E+00 -6.3250E+00 -3.7956E+00 R16 7.3209E-02 1.2282E-02 -3.9264E-03 -2.7496E-02 -8.2491E-02 -1.3745E-01 -1.3736E-01 -8.2341E-02 X.sup.4Y.sup.10 X.sup.2Y.sup.12 X.sup.0Y.sup.12 X.sup.16Y.sup.0 X.sup.14Y.sup.2 X.sup.12Y.sup.4 X.sup.10Y.sup.6 X.sup.8Y.sup.8 R15 -1.2649E+00 -1.8039E-01 3.2338E-02 2.5887E-01 9.0625E-01 1.8120E+00 2.2653E+00 1.8120E+00 R16 -2.7450E-02 -3.9226E-03 6.8211E-04 5.4583E-03 1.9106E-02 3.8233E-02 4.7740E-02 3.8230E-02 X.sup.6Y.sup.10 X.sup.4Y.sup.12 X.sup.2Y.sup.14 X.sup.0Y.sup.16 X.sup.18Y.sup.0 X.sup.16Y.sup.2 X.sup.14Y.sup.4 X.sup.12Y.sup.6 R15 9.0587E-01 2.5843E-01 3.2340E-02 -2.9863E-03 -2.6862E-02 -1.0741E-01 -2.5051E-01 -3.7583E-01 R16 1.9084E-02 5.4408E-03 6.6986E-04 -6.2634E-05 -5.6364E-04 -2.2522E-03 -5.2510E-03 -7.8910E-03 X.sup.10Y.sup.8 X.sup.8Y.sup.10 X.sup.6Y.sup.12 X.sup.4Y.sup.14 X.sup.2Y.sup.16 X.sup.0Y.sup.18 X.sup.20Y.sup.0 X.sup.18Y.sup.2 R15 -3.7581E-01 -2.5092E-01 -1.0736E-01 -2.7143E-02 -3.0706E-03 1.0470E-04 1.0352E-03 4.6028E-03 R16 -7.8922E-03 -5.2547E-03 -2.2758E-03 -5.7182E-04 -6.0808E-05 2.3950E-06 2.3933E-05 1.0736E-04 X.sup.16Y.sup.4 X.sup.14Y.sup.6 X.sup.12Y.sup.8 X.sup.10Y.sup.10 X.sup.8Y.sup.12 X.sup.6Y.sup.14 X.sup.4Y.sup.16 X.sup.2Y.sup.18 R15 1.2348E-02 2.1466E-02 2.5826E-02 2.1665E-02 1.2462E-02 4.6203E-03 1.2040E-03 1.2221E-04 R16 2.8501E-04 5.0026E-04 6.0427E-04 4.9889E-04 2.8755E-04 1.1239E-04 2.6720E-05 2.3405E-06

$$(62) \quad z = \frac{c^2}{1 + \sqrt{1 - (1+k)c^2 r^2}} + \sum_{i=1}^N \text{Math. } B_i E_i(x, y), \quad (2)$$

(63) where k represents a conic coefficient, Bi represents a free-form surface coefficient, c represents the curvature at the center of the optical surface, r represents a vertical distance between the a point on the free-form surface and the optical axis, x represents the x-direction component of r, y represents the y-direction component of r, z represents aspherical depth (a vertical distance between a point on an aspherical surface, having a distance of r from the optic axis, and a surface tangent to a vertex of the aspherical surface on the optic axis)

(64) For convenience, each free-form surface adopts the surface type defined by the extended polynomial shown in the above equation (1). However, the present disclosure is not limited to the free-form surface defined by the polynomial form expressed by the equation (1).

(65) FIG. 2 shows a situation where the RMS spot diameter of the camera optical lens **10** according to the Embodiment 1 is located in a first quadrant. According to FIG. 2, it can be seen that the camera optical lens **10** according to the Embodiment 1 can achieve good imaging quality.

(66) The following Table 10 shows values corresponding to various numerical values in each of Examples 1, 2, and 3 and the parameters already specified in the condition.

(67) As shown in Table 10, the Embodiment 1 satisfies respective condition.

(68) As an example, an entrance pupil diameter ENPD of the camera optical lens **10** is 1.000 mm, the full FOV image height IH (in a diagonal direction) is 6.000 mm, the image height in an x direction is 4.800 mm, the image height in a y direction is 3.600 mm, and the imaging effect is the best in this rectangular area; the FOV in a diagonal direction is 120.00°, the FOV in the x direction is 107.37°, and the FOV in the y direction is 90.94°. The camera optical lens **10** satisfies the design requirements of a wide angle, ultra-thinness, and a large aperture, and its on-axis and off-axis color aberrations is sufficiently corrected, and the camera optical lens **10** has excellent optical characteristics.

Embodiment 2

(69) The Embodiment 2 is basically the same as the Embodiment 1, and the reference signs in the Embodiment 2 are the same as those in the Embodiment 1, and only a difference thereof will be described in the following.

(70) FIG. 3 illustrates a camera optical lens **20** according to Embodiment 2 of the present disclosure. As an example, the object side surface of the fourth lens L4 is convex in a paraxial region.

(71) Table 4, Table 5, and Table 6 show design data of the camera optical lens **20** according to the Embodiment 2 of the present disclosure. Herein, the object side surface and the image side surface of the first lens L1 are free-form surfaces.

(72) TABLE-US-00004 TABLE 4 R d nd vd S1 ∞ d0 = -2.036 R1 -2.534 d1 = 0.585 nd1 1.5444 v1 56.43 R2 7.156 d2 = 0.699 R3 1.856 d3 = 0.341 nd2 1.6610 v2 20.53 R4 2.852 d4 = 0.332 R5 2.741 d5 = 0.485 nd3 1.5444 v3 56.43 R6 -13.486 d6 = 0.074 R7 469.930 d7 = 0.467 nd4 1.5444 v4 56.43 R8 -2.097 d8 = 0.087 R9 -7.325 d9 = 0.240 nd5 1.6800 v5 18.40 R10 7.328 d10 = 0.166 R11 -5.414 d11 = 0.551 nd6 1.5444 v6 56.43 R12 8.796 d12 = 0.110 R13 3.553 d13 = 0.432 nd7 1.5444 v7 56.43 R14 -1.034 d14 = 0.040 R15 1.601 d15 = 0.449 nd8 1.6032 v8 28.29 R16 0.675 d16 = 0.600 R17 ∞ d17 = 0.210 ndg 1.5168 vg 64.17 R18 ∞ d18 = 0.283

(73) Table 5 shows aspherical data of each lens in the camera optical lens **20** according to the Embodiment 2 of the present disclosure.

(74) TABLE-US-00005 TABLE 5 Conic coefficient Aspherical coefficient k A4 A6 A8 A10 A12 R3 -8.4639E-01 7.9902E-02 7.6132E-02 -3.0516E-01 7.1081E-01 8.5920E-02 R4 8.8252E+00 1.1805E-01 -2.0114E-01 1.2779E+00 -3.3283E+00 4.7535E+00 R5 -1.0370E+00 4.9414E-02 3.9619E-02 -7.4779E-02 1.7602E-01 -1.8546E-01 R6 -1.0000E+01 -1.5176E-01 -1.2523E-01 4.1813E-01 -6.5941E-01 1.0676E+00 R7 -1.0000E+01 -1.4095E-01 -2.0156E-01 2.7863E-01 -3.0324E-01 1.4372E+00 R8 1.6323E+00 -5.4607E-02 -4.9926E-01 1.8066E+00 -4.9448E+00 9.1507E+00 R9 -6.7691E+00 -2.4130E-01 -6.1172E-01 2.2136E+00 -5.6503E+00 1.0070E+01 R10 5.3671E+00 -1.3598E-01 -2.3766E-01 8.2709E-01 -1.6011E+00 2.2737E+00 R11 8.5805E+00 -1.1572E-01 1.2450E-01 -5.2484E-02 -5.7360E-01 1.5884E+00 R12 5.7331E+00 -2.0521E-01 -6.2628E-01 7.4980E-01 5.9374E-01 -2.2884E+00 R13 2.6321E+00 3.5302E-01 -7.8180E-01 9.0930E-01 -4.5231E-01 -3.2862E-01 R14 -6.8345E-01 8.4987E-01 -6.9421E-01 8.9527E-01 -1.2806E+00 1.1212E+00 R15 -9.0298E-01 -2.6012E-01 -1.7603E-01 4.9824E-01 -4.2819E-01 1.9668E-01 R16 -3.9349E+00 -1.6064E-01 8.8155E-02 -2.3090E-02 -6.4730E-04 2.3163E-03 Conic coefficient Aspherical coefficient k A14 A16 A18 A20 / R3 -8.4639E-01 -3.0219E+00 5.4904E+00 -4.1906E+00 1.1724E+00 / R4 8.8252E+00 -2.5760E+00 -4.1313E-01 0.0000E+00 0.0000E+00 / R5 -1.0370E+00 0.0000E+00 0.0000E+00 0.0000E+00 / R6 -1.0000E+01 -5.2971E-01 -2.2791E-01 0.0000E+00 0.0000E+00 / R7 -1.0000E+01 -3.0026E+00 3.2270E+00 -1.3886E+00 0.0000E+00 / R8 1.6323E+00 -1.0116E+01 5.7364E+00 -1.1050E+00 0.0000E+00 / R9 -6.7691E+00 -1.1151E+01 6.4707E+00 -1.4169E+00 0.0000E+00 / R10 5.3671E+00 -2.0829E+00 1.1506E+00 -3.5217E-01 4.6102E-02 / R11 8.5805E+00 -1.7373E+00 9.5540E-01 -2.5959E-01 2.6740E-02 / R12 5.7331E+00 2.6139E+00 -1.5614E+00 4.8978E-01 -6.3442E-02 / R13 2.6321E+00 5.9154E-01 -3.3780E-01 8.8760E-02 -9.0187E-03 / R14 -6.8345E-01 -5.7465E-01 1.7161E-01 -2.7762E-02 1.8867E-03 / R15 -9.0298E-01 -5.2858E-02 8.2421E-03 -6.8086E-04 2.2519E-05 / R16 -3.9349E+00 -7.2536E-04 1.1226E-04 -9.1085E-06 3.1084E-07 /

(75) Table 6 shows the free-form surface data in the camera optical lens **20** according to the Embodiment 2 of the present disclosure.

(76) TABLE-US-00006 TABLE 6 Free-form surface coefficient k X.sup.4Y.sup.0 X.sup.2Y.sup.2 X.sup.0Y.sup.4 X.sup.6Y.sup.0 X.sup.4Y.sup.2 X.sup.2Y.sup.4 X.sup.0Y.sup.6 R1 -2.5000E+01 7.9055E-02 1.5833E-01 7.9158E-02 -4.1266E-02 -1.2412E-01 -1.2398E-01 -4.1420E-02 R2 -5.7138E+00 2.9250E-01 5.8695E-01 2.9275E-01 -3.1533E-01 -9.5176E-01 -9.5121E-01 -3.1559E-01 X.sup.8Y.sup.0 X.sup.6Y.sup.2 X.sup.4Y.sup.4 X.sup.2Y.sup.6 X.sup.0Y.sup.8 X.sup.10Y.sup.0 X.sup.8Y.sup.2 X.sup.6Y.sup.4 R1 1.7047E-02 6.8320E-02 1.0242E-01 6.8246E-02 1.7090E-02 -5.1818E-03



-2.5922E-02 -5.1811E-02 R2 4.6261E-01 1.8566E+00 2.7816E+00 1.8565E+00 4.6154E-01 -5.7853E-01 -2.8948E+00  
-5.7798E+00 X.sup.4Y.sup.6 X.sup.2Y.sup.8 X.sup.0Y.sup.10 X.sup.12Y.sup.0 X.sup.10Y.sup.2 X.sup.8Y.sup.4 X.sup.6Y.sup.6  
X.sup.6Y.sup.6 R1 -5.1830E-02 -2.5892E-02 -5.1748E-03 1.1241E-03 6.7420E-03 1.6844E-02 2.2475E-02  
1.6852E-02 R2 -5.7876E+00 -2.8951E+00 -5.7759E-01 5.3161E-01 3.1898E+00 7.9578E+00 1.0634E+01  
7.9699E+00 X.sup.4Y.sup.8 X.sup.2Y.sup.10 X.sup.0Y.sup.12 X.sup.14Y.sup.0 X.sup.12Y.sup.2 X.sup.10Y.sup.4 X.sup.8Y.sup.6  
X.sup.6Y.sup.8 R1 6.7366E-03 1.1214E-03 -1.6701E-04 -1.1687E-03 -3.5070E-03 -5.8480E-03 -5.8456E-03  
-3.5079E-03 R2 3.1909E+00 5.3183E-01 -3.3001E-01 -2.3102E+00 -6.9231E+00 -1.1544E+01 -1.1565E+01  
-6.9221E+00 X.sup.4Y.sup.10 X.sup.2Y.sup.12 X.sup.0Y.sup.14 X.sup.16Y.sup.0 X.sup.14Y.sup.2 X.sup.12Y.sup.4  
X.sup.10Y.sup.6 X.sup.8Y.sup.8 R1 -1.1718E-03 -1.6737E-04 1.6072E-05 1.2844E-04 4.5125E-04 8.9916E-04  
1.1266E-03 9.0165E-04 R2 -2.3111E+00 -3.2998E-01 1.3015E-01 1.0405E+00 3.6438E+00 7.2854E+00 9.1164E+00  
7.2842E+00 X.sup.6Y.sup.10 X.sup.4Y.sup.12 X.sup.2Y.sup.14 X.sup.0Y.sup.16 X.sup.18Y.sup.0 X.sup.16Y.sup.2  
X.sup.14Y.sup.4 X.sup.12Y.sup.6 R1 4.4933E-04 1.2917E-04 1.6148E-05 -9.0478E-07 -8.1033E-06 -3.2440E-05  
-7.5897E-05 -1.1298E-04 R2 3.6424E+00 1.0400E+00 1.3000E-01 -2.9087E-02 -2.6152E-01 -1.0380E+00  
-2.4586E+00 -3.6567E+00 X.sup.10Y.sup.8 X.sup.8Y.sup.10 X.sup.6Y.sup.12 X.sup.4Y.sup.14 X.sup.2Y.sup.16  
X.sup.0Y.sup.18 X.sup.20Y.sup.0 X.sup.18Y.sup.2 R1 -1.1353E-04 -7.6328E-05 -3.1815E-05 -7.8638E-06 -9.0016E-07  
2.2886E-08 2.2750E-07 9.6673E-07 R2 -3.6396E+00 -2.4451E+00 -1.0501E+00 -2.6055E-01 -2.9234E-02  
2.7652E-03 2.7771E-02 1.1895E-01 X.sup.16Y.sup.4 X.sup.14Y.sup.6 X.sup.12Y.sup.8 X.sup.10Y.sup.10 X.sup.8Y.sup.12  
X.sup.6Y.sup.14 X.sup.4Y.sup.16 X.sup.2Y.sup.18 R1 2.7824E-06 4.5967E-06 5.4493E-06 4.9898E-06 2.4797E-06  
1.0078E-06 1.6508E-07 2.3017E-08 R2 3.3903E-01 5.7717E-01 6.8803E-01 5.6747E-01 3.3625E-01 1.2641E-01  
2.7377E-02 2.8456E-03

(77) FIG. 4 shows a situation where the RMS spot diameter of the camera optical lens **20** according to the Embodiment 2 is located in a first quadrant. According to FIG. 4, it can be seen that the camera optical lens **20** according to the Embodiment 2 can achieve good imaging quality.

(78) As shown in Table 10, the Embodiment 2 satisfies respective condition.

(79) As an example, the entrance pupil diameter ENPD of the camera optical lens **20** is 1.000 mm, the full FOV image height IH (in a diagonal direction) is 6.000 mm, the image height in an x direction is 4.800 mm, the image height in a y direction is 3.600 mm, and the imaging effect is the best in this rectangular area; the FOV in a diagonal direction is 120.00°, the FOV in the x direction is 107.34°, and the FOV in the y direction is 90.86°. The camera optical lens **20** satisfies the design requirements of a wide angle, ultra-thinness, and a large aperture, and its on-axis and off-axis color aberration is sufficiently corrected, and the camera optical lens **20** has excellent optical characteristics.

### Embodiment 3

(80) The Embodiment 3 is basically the same as the Embodiment 1, and the reference signs in the Embodiment 2 are the same as those in the Embodiment 1, and only a difference thereof will be described in the following.

(81) FIG. 5 illustrates a camera optical lens **30** according to Embodiment 3 of the present disclosure. As an example, the image side surface of the first lens L1 is convex in a paraxial region; the third lens L3 has a negative refractive power; the image side surface of the third lens L3 is concave in a paraxial region; and the object side surface of the fourth lens L4 is convex in a paraxial region.

(82) Table 7, Table 8, and Table 9 show design data of the camera optical lens **30** according to the Embodiment 3 of the present disclosure. Herein, the object side surface and the image side surface of the first lens L1 are free-form surfaces.

(83) TABLE-US-00007 TABLE 7 R d nd vd S1  $\infty$  d0 = -2.082 R1 -2.040 d1 = 0.580 nd1 1.5444 v1 56.43 R2 -35.714 d2 = 0.619 R3 2.834 d3 = 0.434 nd2 1.6501 v2 21.44 R4 4.018 d4 = 0.405 R5 4.394 d5 = 0.283 nd3 1.5444 v3 56.43 R6 3.624 d6 = 0.040 R7 2.088 d7 = 0.792 nd4 1.5444 v4 56.43 R8 -2.003 d8 = 0.189 R9 -7.232 d9 = 0.240 nd5 1.6800 v5 18.40 R10 7.687 d10 = 0.167 R11 -9.043 d11 = 0.536 nd6 1.5444 v6 56.43 R12 16.906 d12 = 0.058 R13 5.915 d13 = 0.515 nd7 1.5444 v7 56.43 R14 -1.032 d14 = 0.049 R15 1.422 d15 = 0.403 nd8 1.5661 v8 37.70 R16 0.621 d16 = 0.600 R17  $\infty$  d17 = 0.210 ndg 1.5168 vg 64.17 R18  $\infty$  d18 = 0.278

(84) Table 8 shows aspherical data of each lens in the camera optical lens **30** according to the Embodiment 3 of the present disclosure.

(85) TABLE-US-00008 TABLE 8 Conic coefficient Aspherical coefficient k A4 A6 A8 A10 A12 R3 4.0170E+00 1.1221E-01 -1.7243E-01 9.9811E-01 -3.1960E+00 6.7537E+00 R4 1.2630E+01 1.7437E-01 1.0202E-01 -1.7314E-01 1.8705E+00 -5.1589E+00 R5 5.5673E-01 8.5658E-02 1.6389E-02 2.7386E-02 -8.0102E-02 0.0000E+00 R6 4.9054E+00 -2.0188E-01 5.1102E-01 -1.5968E+00 3.5897E+00 -5.0223E+00 R7 8.5735E-01 -2.0494E-01 3.7812E-01 -9.3193E-01 1.4933E+00 -1.4602E+00 R8 6.6656E-01 -1.5900E-01 8.9520E-02 1.6799E-01 -6.6741E-01 9.8259E-01 R9 4.2072E+01 -4.8327E-01 3.8187E-01 -4.9147E-01 1.4986E+00 -2.9556E+00 R10 7.2285E+00 -2.6643E-01 2.0272E-01 3.2190E-02 -2.5410E-01 3.2213E-01 R11 3.9379E+01 4.7340E-02 -1.1286E-01 2.2824E-01 -3.0850E-01 2.2595E-01 R12 -2.5091E+00 1.9769E-02 -4.1645E-01 1.2463E-01 4.6890E-01 -6.0426E-01 R13 7.4151E+00 2.6331E-01 -5.5503E-01 5.9331E-01 -2.9836E-01 -7.2050E-02 R14 -6.9496E-01 5.3805E-01 -7.2255E-01 1.5851E+00 -2.0944E+00 1.5797E+00 R15 -1.0256E+00 -4.4571E-01 1.2135E-01 1.7726E-01 -2.3107E-01 1.2840E-01 R16 -3.0216E+00 -2.4217E-01 2.0139E-01 -1.1423E-01 4.3717E-02 -1.1258E-02 Conic coefficient Aspherical coefficient k A14 A16 A18 A20 / R3 4.0170E+00 -9.0422E+00 7.3506E+00 -3.2918E+00 6.1453E-01 / R4 1.2630E+01 7.1238E+00 -3.7840E+00 0.0000E+00 0.0000E+00 / R5 5.5673E-01 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 / R6 4.9054E+00 3.9416E+00 -1.4186E+00 0.0000E+00 0.0000E+00 / R7 8.5735E-01 6.5417E-01 1.3833E-01 -1.8948E-01 0.0000E+00 / R8 6.6656E-01 -7.2930E-01 2.5177E-01 0.0000E+00 0.0000E+00 / R9 4.2072E+01 3.2810E+00 -1.9134E+00 4.5524E-01 0.0000E+00 / R10 7.2285E+00 -1.9789E-01 5.7420E-02 -6.2125E-03 0.0000E+00 / R11 3.9379E+01 -7.9901E-02 1.1570E-02 -3.4844E-04 0.0000E+00 / R12 -2.5091E+00 3.2556E-01 -8.4338E-02 8.5918E-03 0.0000E+00 / R13 7.4151E+00 1.6843E-01 -8.6674E-02 2.0223E-02 -1.8422E-03 / R14 -6.9496E-01

-7.1903E-01 1.9708E-01 2.9981E-02 1.9445E-03 / R15 -1.0256E+00 -4.0245E-02 7.3824E-03 -7.4308E-04 3.1825E-05 / R16 -3.0216E+00 1.9164E-03 -2.0663E-04 1.2789E-05 -3.4634E-07 /

(86) Table 9 shows free-form surface data in the camera optical lens **30** according to the Embodiment 3 of the present disclosure.

(87) TABLE-US-00009 TABLE 9 Free-form surface coefficient k X.sup.4Y.sup.0 X.sup.2Y.sup.2 X.sup.0Y.sup.4 X.sup.6Y.sup.0 X.sup.4Y.sup.2 X.sup.2Y.sup.4 X.sup.0Y.sup.6 R1 -1.6083E+01 8.6539E-02 1.7316E-01 8.6607E-02 -4.2581E-02 -1.2776E-01 -1.2780E-01 -4.2622E-02 R2 5.0000E+01 3.3222E-01 6.6550E-01 3.3252E-01 -3.4750E-01 -1.0441E+00 -1.0456E+00 -3.4758E-01 X.sup.8Y.sup.0 X.sup.6Y.sup.2 X.sup.4Y.sup.4 X.sup.2Y.sup.6 X.sup.0Y.sup.8 X.sup.10Y.sup.0 X.sup.8Y.sup.2 X.sup.6Y.sup.4 R1 1.6912E-02 6.7641E-02 1.0149E-01 6.7652E-02 1.6920E-02 -4.9789E-03 -2.4894E-02 -4.9791E-02 R2 5.2214E-01 2.0900E+00 3.1348E+00 2.0928E+00 5.2174E-01 -6.3161E-01 -3.1589E+00 -6.3154E+00 X.sup.4Y.sup.6 X.sup.2Y.sup.8 X.sup.0Y.sup.10 X.sup.12Y.sup.0 X.sup.10Y.sup.2 X.sup.8Y.sup.4 X.sup.6Y.sup.6 X.sup.6Y.sup.6 R1 -4.9784E-02 -2.4892E-02 -4.9782E-03 1.0350E-03 6.2105E-03 1.5526E-02 2.0698E-02 1.5527E-02 R2 -6.3160E+00 -3.1608E+00 -6.3141E-01 5.4529E-01 3.2722E+00 8.1781E+00 1.0908E+01 8.1755E+00 X.sup.4Y.sup.8 X.sup.2Y.sup.10 X.sup.0Y.sup.12 X.sup.14Y.sup.0 X.sup.12Y.sup.2 X.sup.10Y.sup.4 X.sup.8Y.sup.6 X.sup.6Y.sup.8 R1 6.2089E-03 1.0349E-03 -1.4590E-04 -1.0213E-03 -3.0638E-03 -5.1067E-03 -5.1067E-03 -3.0640E-03 R2 3.2727E+00 5.4547E-01 -3.0691E-01 -2.1483E+00 -6.4451E+00 -1.0742E+01 -1.0744E+01 -6.4430E+00 X.sup.4Y.sup.10 X.sup.2Y.sup.12 X.sup.0Y.sup.14 X.sup.16Y.sup.0 X.sup.14Y.sup.2 X.sup.12Y.sup.4 X.sup.10Y.sup.6 X.sup.8Y.sup.8 R1 -1.0217E-03 -1.4600E-04 1.3252E-05 1.0600E-04 3.7103E-04 7.4212E-04 9.2754E-04 7.4216E-04 R2 -2.1480E+00 -3.0693E-01 1.0396E-01 8.3148E-01 2.9114E+00 5.8198E+00 7.2780E+00 5.8226E+00 X.sup.6Y.sup.10 X.sup.4Y.sup.12 X.sup.2Y.sup.14 X.sup.0Y.sup.16 X.sup.18Y.sup.0 X.sup.16Y.sup.2 X.sup.14Y.sup.4 X.sup.12Y.sup.6 R1 3.7095E-04 1.0611E-04 1.3254E-05 -6.9943E-07 -6.2984E-06 -2.5198E-05 -5.8760E-05 -8.8151E-05 R2 2.9121E+00 8.3098E-01 1.0391E-01 -1.8719E-02 -1.6853E-01 -6.7362E-01 -1.5740E+00 -2.3583E+00 X.sup.10Y.sup.8 X.sup.8Y.sup.10 X.sup.6Y.sup.12 X.sup.4Y.sup.14 X.sup.2Y.sup.16 X.sup.0Y.sup.18 X.sup.20Y.sup.0 X.sup.18Y.sup.2 R1 -8.8141E-05 -5.8753E-05 -2.5191E-05 -6.2684E-06 -6.9546E-07 1.6266E-08 1.6350E-07 7.3446E-07 R2 -2.3582E+00 -1.5731E+00 -6.7397E-01 -1.6869E-01 -1.8733E-02 1.3166E-03 1.3228E-02 5.9098E-02 X.sup.16Y.sup.4 X.sup.14Y.sup.6 X.sup.12Y.sup.8 X.sup.10Y.sup.10 X.sup.8Y.sup.12 X.sup.6Y.sup.14 X.sup.4Y.sup.16 X.sup.2Y.sup.18 R1 1.9581E-06 3.4229E-06 4.1109E-06 3.4218E-06 1.9529E-06 7.3358E-07 1.5860E-07 1.5820E-08 R2 1.5876E-01 2.7843E-01 3.3157E-01 2.7640E-01 1.5804E-01 5.9097E-02 1.3438E-02 1.3275E-03

(88) FIG. 6 shows a situation where the RMS spot diameter of the camera optical lens **30** according to the Embodiment 3 is located in a first quadrant. According to FIG. 4, it can be seen that the camera optical lens **30** according to the Embodiment 3 can achieve good imaging quality.

(89) The following Table 10 lists the respective numerical value corresponding to each condition in this embodiment according to the above-mentioned condition. Obviously, the imaging optical system according to this embodiment satisfies the above-mentioned condition.

(90) As an example, the entrance pupil diameter ENPD of the camera optical lens **30** is 1.000 mm, the full FOV image height IH (in a diagonal direction) is 6.000 mm, the image height in an x direction is 4.800 mm, the image height in a y direction is 3.600 mm, and the imaging effect is the best in this rectangular area; the FOV in a diagonal direction is 119.99°, the FOV in the x direction is 107.17°, and the FOV in the y direction is 89.67°. The camera optical lens **30** satisfies the design requirements of a wide angle, ultra-thinness, and a large aperture, and its on-axis and off-axis color aberration is sufficiently corrected, and the camera optical lens **20** has excellent optical characteristics.

(91) TABLE-US-00010 TABLE 10 Parameters and condition expression Embodiment 1 Embodiment 2 Embodiment 3 f2/f1 -1.912 -2.089 -3.215 R9/R10 -1.919 -1.000 -0.941 d10/d12 2.435 1.509 2.879 f 1.800 1.800 1.800 f1 -3.399 -3.355 -3.986 f2 6.498 7.008 12.813 f3 4.457 4.216 -43.478 f4 3.440 3.825 2.009 f5 -5.703 -5.300 -5.391 f6 -4.651 -6.053 -10.709 f7 1.508 1.517 1.653 f8 -2.287 -2.353 -2.371 FNO 1.800 1.800 1.800 TTL 6.151 6.151 6.398 FOV 120.00° 120.00° 119.99° IH 6.000 6.000 6.000

(92) The above description merely illustrates some embodiments of the present disclosure. It should be noted that those skilled in the art may make improvements without departing from a creative concept of the present disclosure, and all these improvements shall fall into a protection scope of the present disclosure.

## Claims

1. A camera optical lens, comprising, from an object side to an image side, a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens; wherein the first lens has a negative refractive power, the second lens has a positive refractive power, the fourth lens has a positive refractive power, the fifth lens has a negative refractive power, the sixth lens has a negative refractive power, the seventh lens has a positive refractive power, and the eighth lens has a negative refractive power, wherein at least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens, the sixth lens, the seventh lens, or the eighth lens has a free-form surface, and the camera optical lens satisfies:

$$-3.50 \leq f2/f1 \leq -1.50;$$

$$-2.00 \leq R9/R10 \leq -0.80; \text{ and}$$

$2.435 \leq d10/d12 \leq 3.00$ , where f1 denotes a focal length of the first lens, f2 denotes a focal length of the second lens, R9 denotes a central curvature radius of an object side surface of the fifth lens, R10 denotes a central curvature radius of an image side surface of the fifth lens, d10 denotes an on-axis distance from the image side surface of the fifth lens to an object side surface of the sixth lens, and d12 denotes an on-axis distance from an image side surface of the sixth lens to an object side surface of the seventh lens.

2. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:

$-4.43 \leq f_1/f \leq -1.24$ ;  
 $-2.24 \leq (R_1+R_2)/(R_1-R_2) \leq -0.28$ ; and  
 $0.05 \leq d_1/TTL \leq 0.14$ , where  $f$  denotes a focal length of the camera optical lens,  $R_1$  denotes a central curvature radius of an object side surface of the first lens,  $R_2$  denotes a central curvature radius of an image side surface of the first lens,  $d_1$  denotes an on-axis thickness of the first lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

3. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:  
 $1.81 \leq f_2/f \leq 10.68$ ;  
 $-11.57 \leq (R_3+R_4)/(R_3-R_4) \leq -2.79$ ; and  
 $0.03 \leq d_3/TTL \leq 0.10$ , where  $f$  denotes a focal length of the camera optical lens,  $R_3$  denotes a central curvature radius of an object side surface of the second lens,  $R_4$  denotes a central curvature radius of an image side surface of the second lens,  $d_3$  denotes an on-axis thickness of the second lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

4. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:  
 $-48.31 \leq f_3/f \leq 3.71$ ;  
 $-1.57 \leq (R_5+R_6)/(R_5-R_6) \leq 15.62$ ; and  
 $0.02 \leq d_5/TTL \leq 0.12$ , where  $f$  denotes a focal length of the camera optical lens,  $f_3$  denotes a focal length of the third lens,  $R_5$  denotes a central curvature radius of an object side surface of the third lens,  $R_6$  denotes a central curvature radius of an image side surface of the third lens,  $d_5$  denotes an on-axis thickness of the third lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

5. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:  
 $0.56 \leq f_4/f \leq 3.19$ ;  
 $0.01 \leq (R_7+R_8)/(R_7-R_8) \leq 1.51$ ; and  
 $0.04 \leq d_7/TTL \leq 0.19$ , where  $f$  denotes a focal length of the camera optical lens,  $f_4$  denotes a focal length of the fourth lens,  $R_7$  denotes a central curvature radius of an object side surface of the fourth lens,  $R_8$  denotes a central curvature radius of an image side surface of the fourth lens,  $d_7$  denotes an on-axis thickness of the fourth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

6. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:  
 $-6.34 \leq f_5/f \leq -1.96$ ;  
 $-0.06 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 0.47$ ; and  
 $0.02 \leq d_9/TTL \leq 0.07$ , where  $f$  denotes a focal length of the camera optical lens,  $f_5$  denotes a focal length of the fifth lens,  $R_9$  denotes a central curvature radius of the object side surface of the fifth lens,  $R_{10}$  denotes a central curvature radius of the image side surface of the fifth lens,  $d_9$  denotes an on-axis thickness of the fifth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

7. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:  
 $-11.90 \leq f_6/f \leq -1.72$ ;  
 $-0.71 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq -0.16$ ; and  
 $0.04 \leq d_{11}/TTL \leq 0.13$ , where  $f$  denotes a focal length of the camera optical lens,  $f_6$  denotes a focal length of the sixth lens,  $R_{11}$  denotes a central curvature radius of an object side surface of the sixth lens,  $R_{12}$  denotes a central curvature radius of an image side surface of the sixth lens,  $d_{11}$  denotes an on-axis thickness of the sixth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

8. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:  
 $0.42 \leq f_7/f \leq 1.38$ ;  
 $0.25 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq 1.05$ ; and  
 $0.04 \leq d_{13}/TTL \leq 0.12$ , where  $f$  denotes a focal length of the camera optical lens,  $f_7$  denotes a focal length of the seventh lens,  $R_{13}$  denotes a central curvature radius of an object side surface of the seventh lens,  $R_{14}$  denotes a central curvature radius of an image side surface of the seventh lens,  $d_{13}$  denotes an on-axis thickness of the seventh lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

9. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies:  
 $-2.63 \leq f_8/f \leq -0.85$ ;  
 $1.14 \leq (R_{15}+R_{16})/(R_{15}-R_{16}) \leq 3.83$ ; and  
 $0.03 \leq d_{15}/TTL \leq 0.11$ , where  $f$  denotes a focal length of the camera optical lens,  $f_8$  denotes a focal length of the eighth lens,  $R_{15}$  denotes a central curvature radius of an object side surface of the eighth lens,  $R_{16}$  denotes a central curvature radius of an image side surface of the eighth lens,  $d_{15}$  denotes an on-axis thickness of the eighth lens, and  $TTL$  denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

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